



一. 选择 无答案 $\int_0^1 dx \int_x^{\sqrt{x}} f(x,y) dy$ 或 $\int_0^1 dy \int_{y^2}^{\sqrt{y}} f(x,y) dx$.

1. C 2. ? 3. B 4. C 5. C 6. D 7. B 8. D 9. C

10. B 11. A 12. C 13. B 14. A

二. 计算

1. $\iint_D \sin(x^2 y^2) d\sigma$ 关于 x 是奇函数, D 关于 y 轴对称

$\therefore$ 原式 = 0

2. $\iint_D (x-y^2) dx dy = \int_0^{\pi} dx \int_0^{\sin x} (x-y^2) dy$

$= \int_0^{\pi} dx \left[xy - \frac{1}{3} y^3 \right]_0^{\sin x}$

$= \int_0^{\pi} (x \sin x - \frac{1}{3} (\sin x)^3) dx$

$= \int_0^{\pi} x \sin x dx - \frac{1}{3} \int_0^{\pi} (\sin x)^3 dx$

$= -x \cos x \Big|_0^{\pi} + \frac{1}{3} \int_0^{\pi} (1 - \cos^2 x) d(\cos x)$

$= -2 \cos \pi + \sin \pi \Big|_0^{\pi} + \frac{1}{3} (\cos x - \frac{1}{3} \cos^3 x - 1 + \frac{1}{3})$

$= \pi + \frac{1}{3} (\frac{2}{3} - 2) = \pi - \frac{4}{9}$

3. $\iint_D xy dx dy = \int_0^2 dx \int_0^{x^2} xy dy$

$= \int_0^2 x dx \left(\frac{1}{2} y^2 \Big|_0^{x^2} \right)$

$= \int_0^2 x dx \cdot \frac{1}{2} x^4$

$= \int_0^2 \frac{1}{2} x^5 dx$

$= \frac{1}{12} x^6 \Big|_0^2$

$= \frac{2^6}{12} = \frac{16}{3}$

4. $\iint_D \sqrt{x} y dx dy = \int_0^4 dx \int_x^{\sqrt{x}} \sqrt{x} y dy$

$= \int_0^4 dx \sqrt{x} \left[\frac{1}{2} y^2 \Big|_x^{\sqrt{x}} \right]$

$= \int_0^4 \sqrt{x} \cdot \frac{1}{2} x^2 dx$

$= \frac{1}{2} \int_0^4 x^{\frac{5}{2}} dx$

$= \frac{1}{2} \times \frac{2}{7} x^{\frac{7}{2}} \Big|_0^4$

$= \frac{1}{7} \times 4^{\frac{7}{2}} = \frac{384}{7}$

5. $\iint_D xy dx dy = \int_1^2 dx \int_x^{2x} xy dy$

$= \int_1^2 dx \cdot x \cdot \left(\frac{1}{2} y^2 \Big|_x^{2x} \right)$

$= \int_1^2 x^2 \cdot x dx$

$= \frac{1}{4} x^4 \Big|_1^2 = 4 - \frac{1}{4}$

6. $\iint_D \cos(x+y) dx dy = \int_0^{\pi} dy \int_0^y \cos(x+y) dx$

$= \int_0^{\pi} dy \left[\sin(x+y) \Big|_0^y \right]$

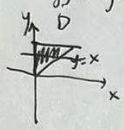
$= \int_0^{\pi} (\sin 2y - \sin y) dy$

$= \int_0^{\pi} \sin 2y dy - \int_0^{\pi} \sin y dy$

$= \frac{1}{2} \cos 2y \Big|_0^{\pi} + \cos y \Big|_0^{\pi}$

$= -\frac{1}{2} (\cos 2\pi - \cos 0) + (\cos \pi - \cos 0)$

$= -1 - 1 = -2$

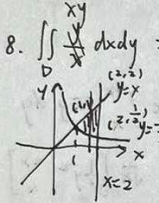
$$7. \iint_D (x+y) dx dy = \int_0^1 dy \int_0^y (x+y) dx$$


$$= \int_0^1 dy \left[\frac{1}{2}x^2 + xy \right]_0^y$$

$$= \int_0^1 \left(\frac{1}{2}y^2 + y^2 \right) dy$$

$$= \frac{3}{2} \int_0^1 y^2 dy$$

$$= \frac{3}{2} \times \frac{1}{3} y^3 \Big|_0^1 = \frac{1}{2}$$

$$8. \iint_D \frac{xy}{x^2} dx dy = \int_1^2 dx \int_{\frac{1}{x}}^x xy dy$$


$$= \int_1^2 dx \left[\frac{1}{2}xy^2 \Big|_{\frac{1}{x}}^x \right]$$

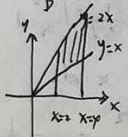
$$= \int_1^2 dx \left[\frac{1}{2}x^3 - \frac{1}{2x} \right]$$

$$= \frac{1}{2} \int_1^2 (x^3 - x^{-1}) dx$$

$$= \frac{1}{2} \left(\frac{1}{4}x^4 - \ln x \right) \Big|_1^2$$

$$= \frac{1}{2} \left[4 - \ln 2 - \frac{1}{4} + 0 \right]$$

$$= \frac{15}{8} - \frac{\ln 2}{2}$$

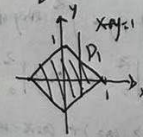
$$9. \iint_D \frac{y}{x} dx dy = \int_2^4 dx \int_x^{2x} \frac{y}{x} dy$$


$$= \int_2^4 dx \left(\frac{y^2}{2x} \Big|_x^{2x} \right)$$

$$= \int_2^4 dx \left(\frac{4x^2 - x^2}{2x} \right)$$

$$= \int_2^4 \frac{3}{2}x dx$$

$$= \frac{3}{4}x^2 \Big|_2^4 = \frac{3}{4}(16-4) = 9$$

$$10. \iint_D |y| dx dy = 4 \int_0^1 y dx dy$$


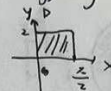
$$= 4 \int_0^1 dy \int_0^{1-y} y dy$$

$$= 4 \int_0^1 dy \left(\frac{1}{2}y^2 \Big|_0^{1-y} \right)$$

$$= 4 \int_0^1 dy (xy) \Big|_0^{1-y}$$

$$= 4 \int_0^1 y(1-y) dy = 4 \left(\frac{1}{2}y^2 - \frac{1}{3}y^3 \right) \Big|_0^1$$

$$= 4 \times \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{2}{3}$$

$$11. \iint_D x^2 y \cos(xy^2) dx dy = \int_0^{\frac{\pi}{2}} dx \int_0^2 \cos(xy^2) x^2 y dy$$


$$= \frac{1}{2x} \Big|_0^{\frac{\pi}{2}} x^2 dx \int_0^2 \cos(xy^2) d(xy^2)$$

$$= \frac{1}{2x} \Big|_0^{\frac{\pi}{2}} x^2 dx \sin(xy^2) \Big|_0^2$$

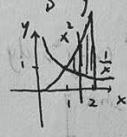
$$= \int_0^{\frac{\pi}{2}} \frac{x}{2} \sin 4x dx$$

$$= \int_0^{\frac{\pi}{2}} -\frac{x}{8} d(\cos 4x)$$

$$= -\frac{x}{8} \cos 4x \Big|_0^{\frac{\pi}{2}} + \left(\frac{1}{32} \right) \cos 4x \Big|_0^{\frac{\pi}{2}}$$

$$= -\frac{\pi}{16} + \left(-\frac{1}{32} \right) \sin 4x \Big|_0^{\frac{\pi}{2}}$$

$$= -\frac{\pi}{16}$$

$$12. \iint_D \frac{x^2}{y^2} dx dy = \int_1^2 dx \int_{\frac{1}{x}}^x \frac{x^2}{y^2} dy$$


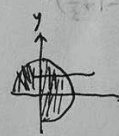
$$= \int_1^2 x^2 dx \left(-y^{-1} \Big|_{\frac{1}{x}}^x \right)$$

$$= \int_1^2 x^2 dx \left[-\frac{1}{x^2} + x \right]$$

$$= \int_1^2 (x^3 - 1) dx$$

$$= \frac{1}{4}x^4 - x \Big|_1^2 = 4 - 2 - \frac{1}{4} + 1$$

$$= \frac{11}{4}$$

$$13. \iint_D xy^2 dx dy = \int_{-2}^2 dy \int_0^{\sqrt{4-y^2}} xy^2 dx$$


$$= \int_{-2}^2 y^2 dy \left(\frac{1}{2}x^2 \Big|_0^{\sqrt{4-y^2}} \right)$$

$$= \int_{-2}^2 y^2 \cdot \frac{1}{2}(x-y^2) dy$$

$$= \int_{-2}^2 (2y^2 - \frac{1}{2}y^4) dy$$

$$= \left(\frac{2}{3}y^3 - \frac{1}{10}y^5 \right) \Big|_{-2}^2$$

$$= 2 \times \left(\frac{2}{3} \times 8 - \frac{1}{10} \times 32 \right)$$

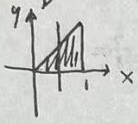
$$= 2 \times \left(\frac{16}{3} - \frac{16}{5} \right) = \frac{16}{15} \times 4 = \frac{64}{15}$$



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$$\int_0^2 x d(\ln x) = \left. x \ln x \right|_0^2 - \int_0^2 1 dx = 2 \ln 2 - 2$$

14. $\iint_D \frac{1}{1+x^4} dx dy = \int_0^1 dx \int_0^x \frac{1}{1+x^4} dy$



$$= \int_0^1 \frac{x}{1+x^4} dx$$

$$= \frac{1}{2} \int_0^1 \frac{1}{1+x^2} dx$$

$$= \frac{1}{2} \arctan(x^2) \Big|_0^1$$

$$= \frac{1}{2} \arctan 1 = \frac{\pi}{8}$$

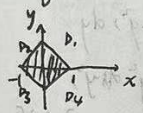
3. $\iint_D \arctan \frac{y}{x} dx dy = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r dr$



$$= \int_0^{\frac{\pi}{2}} d\theta \left. \frac{1}{2} r^2 \right|_0^1$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} d\theta = \frac{1}{2} \theta \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{4}$$

15. $\iint_D e^{x+y} d\sigma = \int_{-1}^0 e^x dx \int_{x-1}^{x+1} e^y dy$

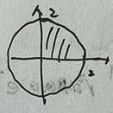


$$= \int_{-1}^0 e^x dx \int_{x-1}^{x+1} e^y dy$$

$$= \int_{-1}^0 (e^{2x+1} - e^{-1}) dx = \int_0^1 (e^{2x} - e^{-1}) dx$$

$$= e - \frac{1}{e}$$

4. $\iint_D \ln(1+x^2+y^2) dx dy$



$$= \int_0^{\frac{\pi}{2}} d\theta \int_0^2 \ln(1+r^2) r dr$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \int_0^2 \ln(1+r^2) d(r^2)$$

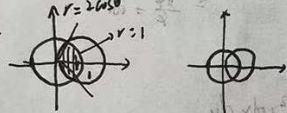
$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \left[(r^2+1) \ln(r^2+1) - r^2 \right]_0^2$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta [5 \ln 5 - 4]$$

$$= \frac{\pi}{4} (5 \ln 5 - 4) = \frac{5}{4} \ln 5 \cdot \pi - \pi$$

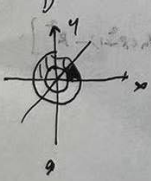
三. 计算积分

1. $\iint_D f(x,y) dx dy = \int_0^{\frac{\pi}{2}} d\theta \int_{2\cos\theta}^1 f(r\cos\theta, r\sin\theta) r dr$



$$= \int_0^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} f(r\cos\theta, r\sin\theta) r dr + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta \int_{2\cos\theta}^1 f(r\cos\theta, r\sin\theta) r dr$$

2. $\iint_D \arctan \frac{y}{x} dx dy = \int_0^{\frac{\pi}{4}} d\theta \int_1^2 \arctan(\tan\theta) r dr$



$$= \int_0^{\frac{\pi}{4}} d\theta \int_1^2 \arctan(\tan\theta) r dr$$

$$= \int_0^{\frac{\pi}{4}} d\theta \left. \frac{1}{2} r^2 \right|_1^2$$

$$= \int_0^{\frac{\pi}{4}} \frac{3}{2} d\theta = \frac{3}{4} \theta \Big|_0^{\frac{\pi}{4}} = \frac{3\pi}{16}$$

5. $\iint_D (x^2+y^2) dx dy = \int_0^{\frac{\pi}{2}} d\theta \int_{2\cos\theta}^{2\sin\theta} r^3 dr$



$$= \int_0^{\frac{\pi}{2}} d\theta \left(\frac{1}{4} r^4 \right)_{2\cos\theta}^{2\sin\theta}$$

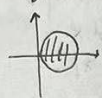
$$= \int_0^{\frac{\pi}{2}} \frac{16\sin^4\theta - 16\cos^4\theta}{4} d\theta$$

$$= \int_0^{\frac{\pi}{2}} 4(\sin^4\theta - \cos^4\theta) d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} (\sin^2\theta - \cos^2\theta) d\theta = 4 \int_0^{\frac{\pi}{2}} -\cos 2\theta d\theta$$

$$= 4 \left(-\frac{1}{2} \sin 2\theta \right) \Big|_0^{\frac{\pi}{2}} = 0$$

$$6. \iint_D \sqrt{x^2+y^2} dx dy = \int_0^{\pi} d\theta \int_0^{2\cos\theta} r^2 dr$$



$$= \int_0^{\pi} d\theta \left[\frac{1}{3} r^3 \right]_0^{2\cos\theta}$$

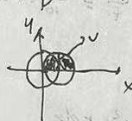
$$= \int_0^{\pi} \frac{8}{3} \cos^3\theta d\theta$$

$$= \frac{8}{3} \int_0^{\pi} (1 - \sin^2\theta) d(\sin\theta)$$

$$= \frac{8}{3} \left(\sin\theta - \frac{1}{3} \sin^3\theta \right) \Big|_0^{\pi}$$

$$= 0$$

$$7. \iint_D xy dx dy = \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta \int_0^{2\cos\theta} r^2 dr$$



$$= \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta \left[\frac{1}{3} r^3 \right]_0^{2\cos\theta}$$

$$= \frac{1}{3} \int_0^{\frac{\pi}{2}} \sin\theta \cos^3\theta d\theta$$

$$= \frac{1}{3} \left[\frac{1}{2} \cos^2\theta - \frac{1}{8} \cos^4\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{16}$$

$$8. \iint_D xy e^{-x^2-y^2} dx dy = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2 \sin\theta \cos\theta e^{-r^2} r dr$$



$$= \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta \int_0^1 r^3 e^{-r^2} dr$$

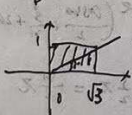
$$= \int_0^{\frac{\pi}{2}} \frac{e^{-2} - 1}{4} \sin\theta \cos\theta d\theta$$

$$= \frac{e^{-2} - 1}{16} \int_0^{\frac{\pi}{2}} \sin 2\theta d(2\theta)$$

$$= \frac{e^{-2} - 1}{16} \left[-\cos 2\theta \right]_0^{\frac{\pi}{2}}$$

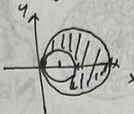
$$= \frac{e^{-2} - 1}{16} [\cos 0 - \cos \pi] = \frac{e^{-2} - 1}{8}$$

$$9. \iint_D f(x,y) dx dy = \int_0^{\frac{\pi}{3}} d\theta \int_0^{\frac{1}{\cos\theta}} f(r\cos\theta, r\sin\theta) r dr$$



$$+ \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\theta \int_0^{\frac{1}{\sin\theta}} f(r\cos\theta, r\sin\theta) r dr$$

$$10. \iint_D x dx dy = \int_0^{\pi} d\theta \int_{\cos\theta}^{2\cos\theta} r \cos\theta r dr$$



$$= \int_0^{\pi} \cos\theta d\theta \int_{\cos\theta}^{2\cos\theta} r^2 dr$$

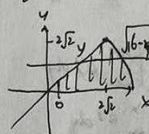
$$= \int_0^{\pi} \cos\theta d\theta \left[\frac{1}{3} r^3 \right]_{\cos\theta}^{2\cos\theta}$$

$$= \int_0^{\pi} \frac{1}{3} \cos\theta (7\cos^3\theta) d\theta$$

$$= \frac{1}{3} \int_0^{\pi} 7\cos^4\theta d\theta$$

$$= \frac{7}{3} \int_0^{\pi} \cos^4\theta d\theta = \frac{7}{8} \pi$$

$$11. \iint_D xy dx dy = \int_0^{2\sqrt{2}} dy \int_y^{\sqrt{6-y^2}} xy dx$$



$$= \int_0^{2\sqrt{2}} y dy \left[\frac{1}{2} x^2 \right]_y^{\sqrt{6-y^2}}$$

$$= \int_0^{2\sqrt{2}} y (8-y^2) dy$$

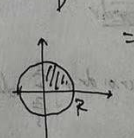
$$= \frac{1}{2} \int_0^{2\sqrt{2}} (8-y^2) dy$$

$$= \frac{1}{2} \left[8y - \frac{1}{3} y^3 \right]_0^{2\sqrt{2}}$$

$$= -\frac{1}{2} \times \left[0 - \frac{1}{3} \times 8^2 \right]$$

$$= \frac{64}{3} = \text{cb}$$

$$12. \iint_D \ln(x^2+y^2) dx dy$$



$$= \int_0^{\frac{\pi}{2}} d\theta \int_0^R \ln(r^2) r dr$$

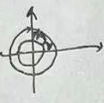
$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \int_0^R \ln(r^2) d(r^2)$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \left[(r^2+1) \ln(r^2+1) - \frac{1}{2} r^2 \right]_0^R$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \left[(R^2+1) \ln(R^2+1) - \frac{1}{2} R^2 \right]$$

$$= \frac{\pi}{4} \left[(R^2+1) \ln(R^2+1) - \frac{1}{2} R^2 \right]$$

13. $\iint_D \sin \sqrt{x^2+y^2} dx dy$



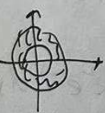
$$\begin{aligned}
 &= \int_0^{\frac{\pi}{2}} d\theta \int_0^1 \sin r \cdot r dr \\
 &= \int_0^{\frac{\pi}{2}} d\theta \left[-\frac{1}{2} r^2 \cos r \right]_0^1 \\
 &= \int_0^{\frac{\pi}{2}} d\theta \left[-\frac{1}{2} \cos 1 + \frac{1}{2} \cos 0 \right] \\
 &= \int_0^{\frac{\pi}{2}} d\theta \left[-\frac{1}{2} \cos 1 + \frac{1}{2} \right] \\
 &= \frac{\pi}{2} \left[-\frac{1}{2} \cos 1 + \frac{1}{2} \right] \\
 &= \frac{\pi}{2} \left[\frac{1}{2} (1 - \cos 1) \right]
 \end{aligned}$$

14. $\iint_D x^2 \sin(x^2+y^2) dx dy$



$$\begin{aligned}
 &= \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2 \cos^2 \theta \sin r^2 \cdot r dr \\
 &= \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \int_0^1 r^3 \sin r^2 dr \\
 &= \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \left[-\frac{1}{4} \cos r^2 \right]_0^1 \\
 &= \int_0^{\frac{\pi}{2}} -\frac{1}{4} (\cos 1 - \cos 0) \cos^2 \theta d\theta \\
 &= \frac{\cos 1 - \cos 0}{4} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \\
 &= \frac{\cos 1 - \cos 0}{4} \int_0^{\frac{\pi}{2}} \frac{1 + \cos 2\theta}{2} d\theta \\
 &= \frac{\cos 1 - \cos 0}{8} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} \\
 &= \frac{\pi}{16} (\cos 1 - \cos 0)
 \end{aligned}$$

15. $\iint_D \frac{(x+iy+1)^2}{x^2+y^2} dx dy$



$$\begin{aligned}
 &= \int_0^{2\pi} d\theta \int_0^1 \frac{(r \cos \theta + i r \sin \theta + 1)^2}{r^2} r dr \\
 &= \int_0^{2\pi} d\theta \int_0^1 \left(r^2 \cos^2 \theta + 2r \cos \theta + \frac{1}{r} + 2r \sin \theta + 2r \sin \theta \cos \theta + \frac{1}{r} \right) dr \\
 &= \int_0^{2\pi} d\theta \left(\frac{1}{2} r^2 \cos^2 \theta + r^2 \sin^2 \theta + \ln r + 2r \cos \theta + 2r \sin \theta \cos \theta + \ln r \right) \Big|_0^1 \\
 &= 2\pi \ln 2 + \frac{1}{2} \times 2\pi + \int_0^{2\pi} \frac{1}{2} \cos^2 \theta d\theta + \int_0^{2\pi} \frac{1}{2} \sin^2 \theta d\theta \\
 &= 2\pi \ln 2 + 3\pi + \int_0^{2\pi} \frac{1}{2} \frac{1 + \cos 2\theta}{2} d\theta + \int_0^{2\pi} \frac{1}{2} \frac{1 - \cos 2\theta}{2} d\theta \\
 &= 2\pi \ln 2 + 3\pi + \frac{1}{4} \times 2\pi \\
 &= 2\pi \ln 2 + \frac{13}{2} \pi
 \end{aligned}$$

16. $\iint_D f(x,y) dx dy = k$

17. $f(x,y) = \sqrt{1-x^2-y^2} - \frac{8}{\pi} k$

$\iint_D f(x,y) dx dy$

$$= \iint_D \sqrt{1-x^2-y^2} dx dy - \frac{8}{\pi} k \iint_D dx dy$$

$\Rightarrow k = \iint_D \sqrt{1-x^2-y^2} dx dy - k$

$\Rightarrow 2k = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r \sqrt{1-r^2} dr$

$$= \frac{\pi}{2} - \frac{2}{3} = \frac{\pi}{6} - \frac{2}{9}$$

$k = \frac{\pi}{12} - \frac{1}{9}$

$\therefore f(x,y) = \sqrt{1-x^2-y^2} - \frac{8}{\pi} \left(\frac{\pi}{12} - \frac{1}{9} \right)$

$$= \sqrt{1-x^2-y^2} - \frac{2}{3} + \frac{8}{9\pi}$$